



SMU

SINGAPORE MANAGEMENT
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G1

IS630 Statistical Thinking for Data Science
Assignment: Applying Descriptive and Inferential Statistics

1. Problem Statement

➤ Background

Property companies undertake various tasks, such as investment, real estate brokerage, and property valuation. They monitor market movements and leverage on real estate analytics to gain insights for higher profitability. For the HDB resale market, where property companies often help with real estate brokerage and valuation tasks, statistical analysis of resale flat transaction data (resale prices, location, lease date, etc.) helps the property company and its real estate agents develop sales strategies, craft a competitive resale pricing for clients and prepare for future planning.

➤ Problem Statement

This analysis aims to explore **buyer demand patterns that may influence a property company's HDB resale brokerage (buy / sell) strategy**. The desired outcome would be to produce recommendations to its real estate agents. We analyse buyer demand pattern in terms of 3 key aspects:

Aspects	Sub Problem Statement
1. Identifying High-Growth Towns	• Which locations show increasing resale prices and transaction volumes? • To identify high-growth locations (at the “town” granularity) and guide both real estate agents to business opportunities, and clients to take advantage of high prices.
2. Relationship between Flat Age and Resale Price	• What is the relationship between flat age and resale price? • To assess any trend between age and resale price, and investigate if there are price differences so as to inform client's valuation strategy
3. Impact of Cooling Measures on Resale Prices & Transactions	• Did government cooling measures introduced in 2021 make resale flats more affordable? • To analyse effectiveness of government policy, anticipate market shifts and adjust strategies.

2. Summary Statistics

➤ Data Cleaning & Transformation

- There are no missing values, data is very complete
- 372 duplicate records found, likely to be invalid given low possibility of transaction with same resale_price, same address and floor range in the same year-month. Treatment: Duplicates removed
- “month” column contained “year” and “month” information. Treatment: Split into 2 separate columns, “year” and “month”
- “town” aggregated into “regions” based on HDB’s [definition](#), to facilitate a more macro analysis
- Magnitude of “resale_price” is affected by flat size. To ensure comparisons of price across property characteristic are fair, we normalized this variable into “price_per_sqm” = resale_price / floor_area_sqm

➤ Preliminary Insights

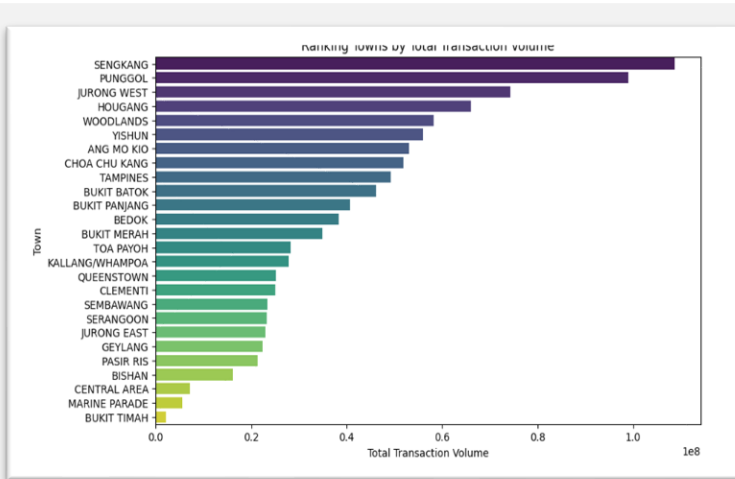
- There are 26 towns, which necessitates the creation of “regions” field
- The oldest flat is 59 years old, and the youngest flat is 4 years old (which may be an anomaly given HDBs cannot be sold in the first 5 years)
- The mean flat size is 97 sqm, with the 50th percentile (aka median) 93 sqm, indicative of a right skewed distribution

	year_month	town	flat_type	block	street_name	storey_range	floor_area_sqm	flat_model	lease_commence_date	resale_price	year	month	regions	price_per_sqm
count	196314	196314	196314	196314	196314	196314	196314	196314	196314	196314	196314	196314	196314	196314
unique	96	26	7	2728	570	17	NaN	21	NaN	NaN	NaN	NaN	5	NaN
top	2024-07	SENGKANG	4 ROOM	2	YISHUN RING RD	04 TO 06	NaN	Model A	NaN	NaN	NaN	NaN	North East	NaN
freq	3036	16194	82917	596	2840	45132	NaN	69040	NaN	NaN	NaN	NaN	52239	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	97.014394	NaN	1996.181724	507995	2020.730666	6.654492	NaN	5288.347182
std	NaN	NaN	NaN	NaN	NaN	NaN	24.036178	NaN	14.142821	177696	2.250329	3.389634	NaN	1504.792333
min	NaN	NaN	NaN	NaN	NaN	NaN	31	NaN	1966	140000	2017	1	NaN	2089.55
25%	NaN	NaN	NaN	NaN	NaN	NaN	82	NaN	1985	376000	2019	4	NaN	4232.88
50%	NaN	NaN	NaN	NaN	NaN	NaN	93	NaN	1996	478000	2021	7	NaN	5000
75%	NaN	NaN	NaN	NaN	NaN	NaN	112	NaN	2011	608000	2023	10	NaN	5959.565
max	NaN	NaN	NaN	NaN	NaN	NaN	366.7	NaN	2021	1588000	2024	12	NaN	15591.4

3. Identifying High-Growth Towns

A. Descriptive Statistics

From the perspective of the property company, towns with high growth potential are those that have high transaction volume and high median resale price per sqm (Median is used as the data is right-skewed). We give each town a combined rank score, and identify the top 5 towns.



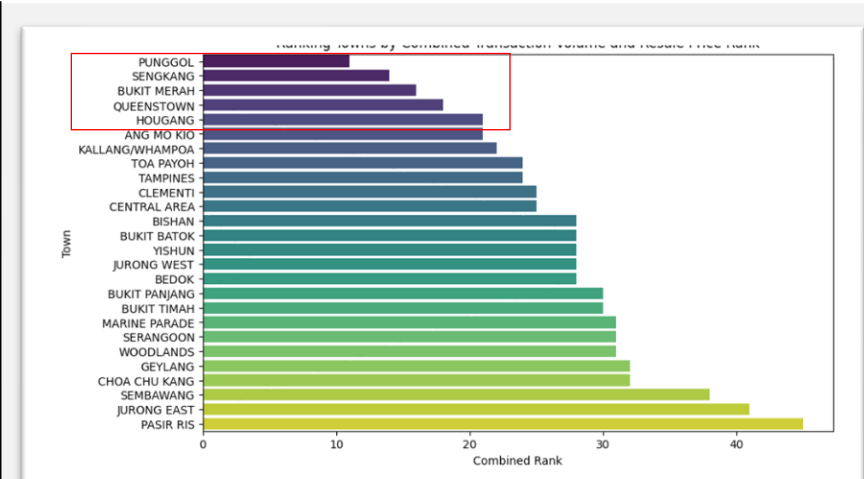
1. Rank of Towns by Transaction Volume:

Sengkang leads in resale transactions, followed by Punggol and Jurong West. Despite being prime locations, Central Area and Marine Parade have lower transaction volumes, likely due to higher prices or fewer number of units.



2. Rank of Towns by Median Resale Price per sqm:

Central Area has the highest prices, followed by Queenstown and Bukit Merah. North and West regions, i.e. Woodlands, Jurong West and Choa Chu Kang have the lowest median resale prices. Punggol and Sengkang balance high transaction volume (see first chart) with mid-range pricing, offering value for money



3. Combined Rank of Towns by Transaction volume and median resale price: Punggol and Sengkang rank highest, showing strong demand and competitive pricing. Towns like Central Area, Clementi and Tampines demonstrate moderate performance for both metrics. Towns lowly ranked (e.g. Pasir Ris, Jurong East, Sembawang) have weakest market performance

Insight: Currently, **Punggol and Sengkang** from the North-East region lead the resale market with a combination of high transaction volumes and competitive resale prices. Next in line for potential growth would be **Bukit Merah and Queenstown** (near Central Region), and **Hougang and Ang Mo Kio** (also in North-East), likely due to their higher in median resale price (i.e. may yield higher sales commission rates) and have moderate transaction volume.

3. Identifying High-Growth Towns

B. Inferential Statistics

Leveraging the insights from the descriptive analysis, we performed hypothesis testing to validate whether **the Top 5 towns** (Punggol, Sengkang, Bukit Merah, Queenstown, Hougang) exhibit significantly higher transaction volumes and resale prices compared to the remaining towns across Singapore.

➤ Approach:

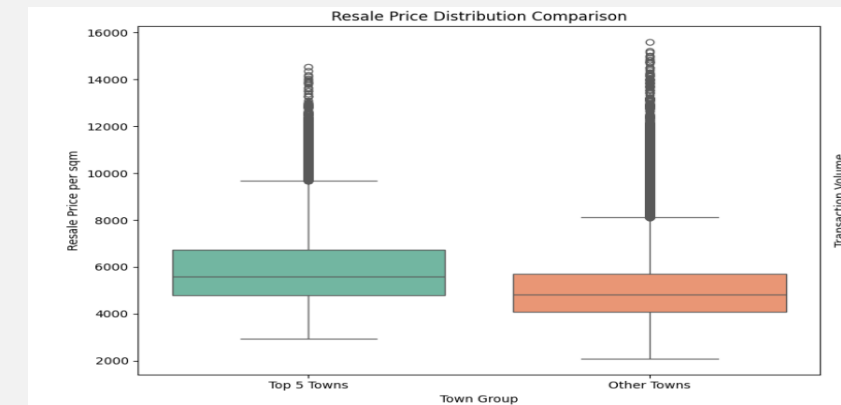
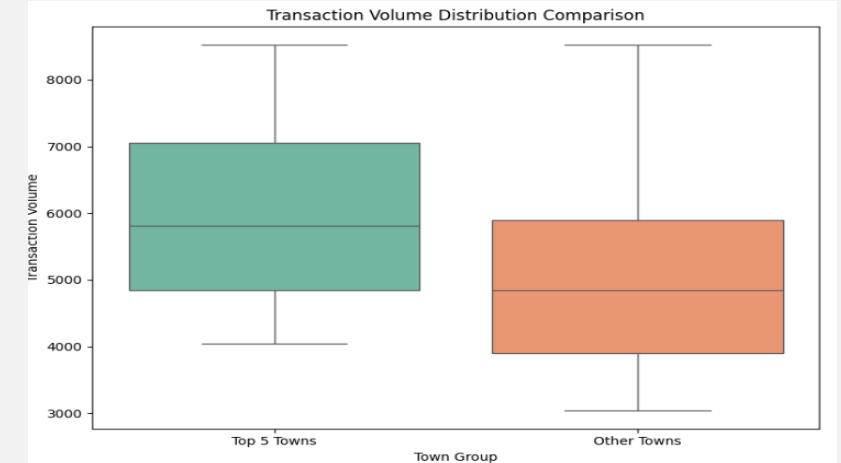
- 2 independent t-tests were performed to determine whether the differences in median resale price per sqm and transaction volume are statistically significant respectively.
- 2 groups were formed for each t-test: The **top 5 towns** VS the **other towns**.
- Null and Alternate Hypothesis:
 - **H0:** There is no significant difference in resale prices (transaction volume) between the top 5 towns and the remaining towns.
 - **H1:** The top 5 towns have significantly higher resale prices (transaction volume) than the remaining towns.

➤ Results:

- For both t-tests, **t-statistic** values are high (**145.36 for Resale Price** and **146.58 for Transaction Volume**), indicating a substantial difference between means of **Top 5 towns** and **Other towns**.
- **p-value** is extremely small (~ 0) < significance level of **0.05**. This further confirms that the difference between the groups is statistically significant, rejecting the null hypothesis (H0).
- Visually, the differences between the 2 groups can also be seen from the box plots (see right)

➤ Conclusion:

- The hypothesis tests confirms that the **Top 5 towns** command **high transactions volume and prices**, making them prime location for property investment.
- Therefore, these top 5 towns have **strong and liquidity and resale value**, enabling them to attract both homeowners and investors.
- These insights may help potential buyers, investors and urban planners better understand the **market dynamics** and anticipate **future high-growth areas**.



4. Relationship between Flat Age & Resale Price / Sqm

A. Descriptive Statistics

Insight: Resale price per sqm decreases with flat age increases, with 2 distinct price drop patterns:

- **Type 1:** Resale price gradually **drops linearly** with age. Towns that exhibit this pattern: Ang Mo Kio, Bedok, Bukit Merah, Clementi, Geylang, Jurong East, Jurong West, Kallang/Whampoa, Punggol, Queenstown, Sengkang, Woodlands, Toa Payoh.
- **Type 2:** Resale price remain stable for certain age ranges, then **drops stepwise** at **key age thresholds**. Towns that exhibit this behaviour: Bishan, Bukit Batok, Bukit Panjang, Central Area, Choa Chu Kang, Hougang, Pasir Ris, Sembawang, Tampines, Yishun.

Note 1: Towns excluded due to insufficient samples to discern pattern: Marine Parade, Bukit Timah, Serangoon.
Note 2: 1-room, 2-room and multi-gen flat types excluded from analysis due to insufficient samples to discern pattern

Strategic Takeaways:

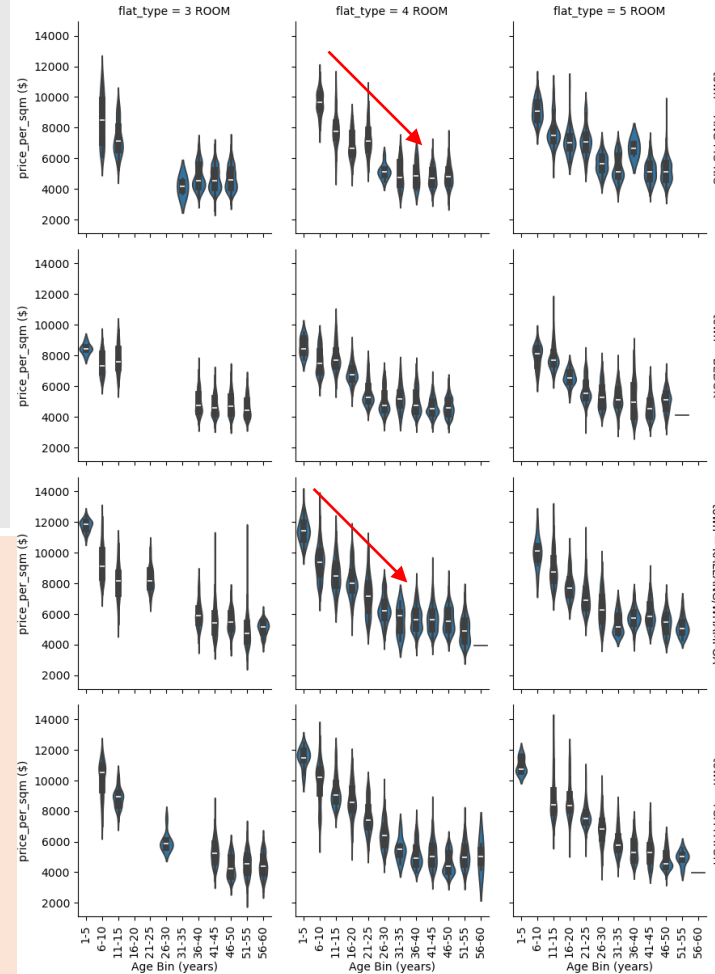
For Sellers/Buyers & Investors:

- Type 1 towns show **more predictable price trends**, good for **long-term investment**.
- Type 2 towns may offer **both buying and selling opportunities**:
 - **Buyers** can **capitalize on affordability** by purchasing after significant price drops.
 - **Sellers** may consider selling **before key age thresholds** to maximize value.

For Property Companies & Urban Planners:

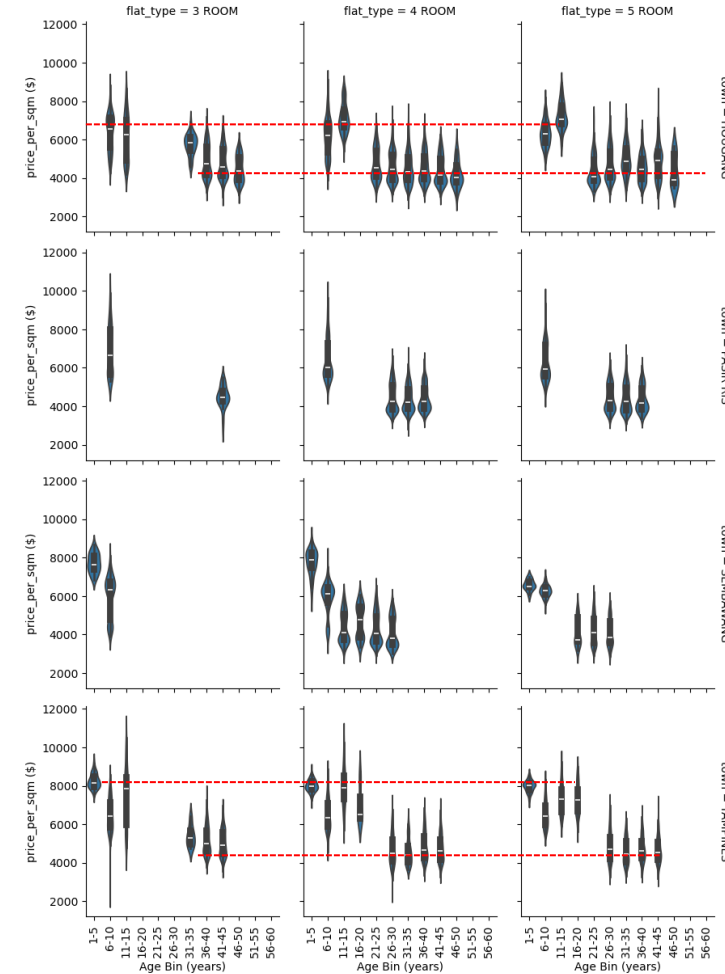
- Better understand these patterns will help them to **refine pricing strategies** for older flats.
- Towns with Type 2 require more **targeted marketing strategies** to get better prices, else they'd stay stagnant after the threshold.

Type 1: Resale price drops linearly with flat age



Examples of towns that exhibit Type 1 behaviour (Ang Mo Kio, Bedok, Kallang/Whampoa, Toa Payoh)

Type 2: Resale price drops stepwise with flat age

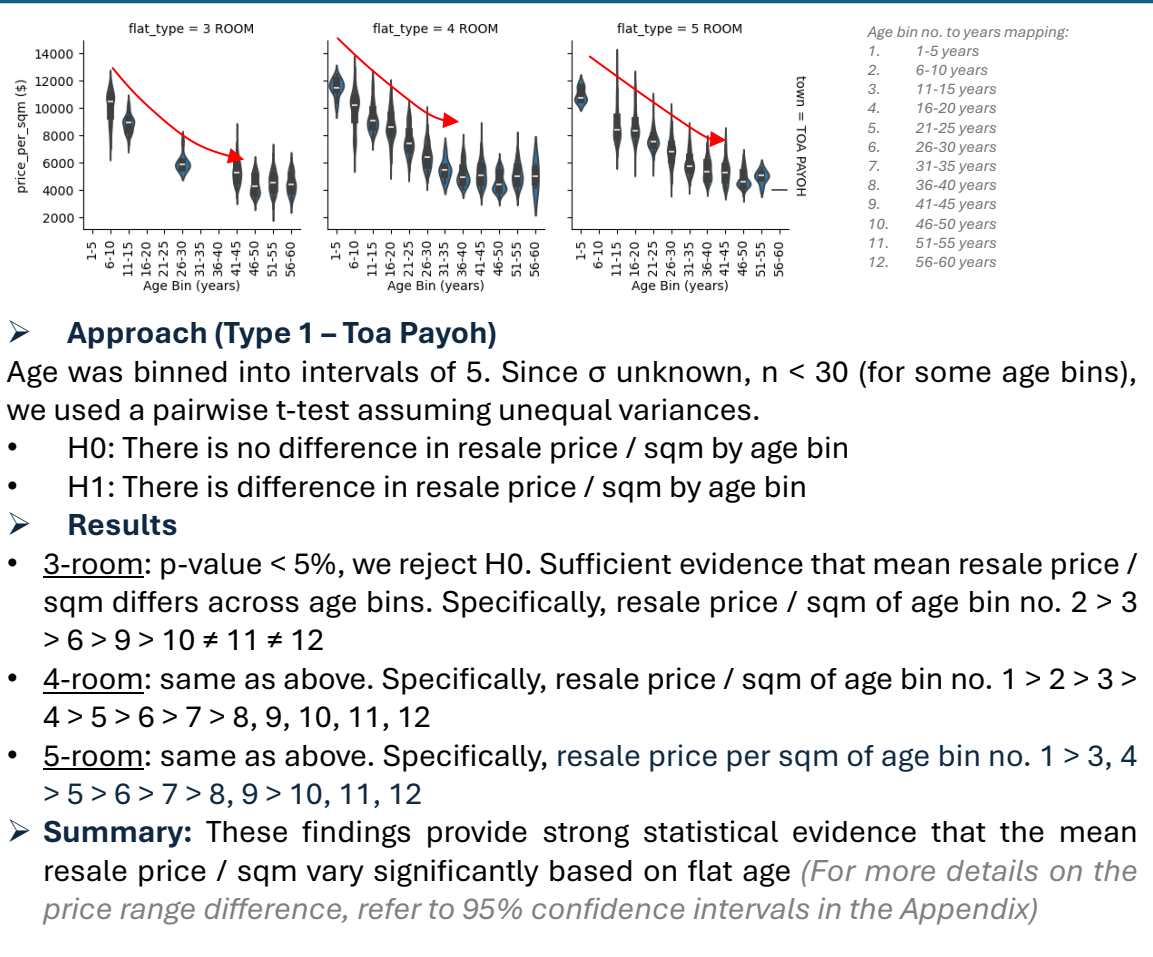


Examples of towns that exhibit Type 2 behaviour (Hougang, Pasir Ris, Sembawang, Tampines)

4. Relationship between Flat Age & Resale Price / Sqm

B. Inferential Statistics

To validate the observed price trends, we performed hypothesis testing on towns for each price behaviour type (Type 1: Toa Payoh; Type 2: Tampines), across different flat types (3 room, 4 room, 5 room)



➤ Approach (Type 1 – Toa Payoh)

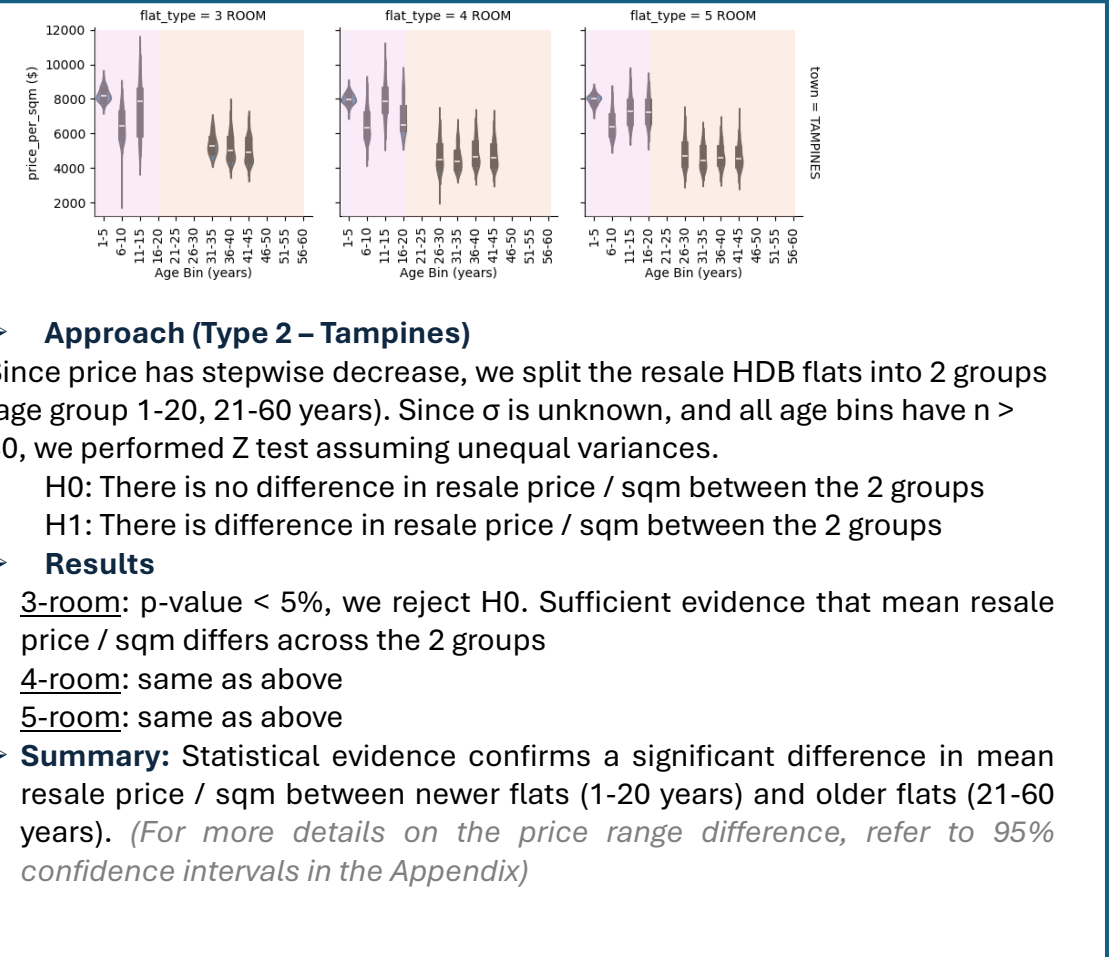
Age was binned into intervals of 5. Since σ unknown, $n < 30$ (for some age bins), we used a pairwise t-test assuming unequal variances.

- H0: There is no difference in resale price / sqm by age bin
- H1: There is difference in resale price / sqm by age bin

➤ Results

- 3-room: p-value < 5%, we reject H0. Sufficient evidence that mean resale price / sqm differs across age bins. Specifically, resale price / sqm of age bin no. 2 > 3 > 6 > 9 > 10 ≠ 11 ≠ 12
- 4-room: same as above. Specifically, resale price / sqm of age bin no. 1 > 2 > 3 > 4 > 5 > 6 > 7 > 8, 9, 10, 11, 12
- 5-room: same as above. Specifically, resale price per sqm of age bin no. 1 > 3, 4 > 5 > 6 > 7 > 8, 9 > 10, 11, 12

➤ **Summary:** These findings provide strong statistical evidence that the mean resale price / sqm vary significantly based on flat age (For more details on the price range difference, refer to 95% confidence intervals in the Appendix)



➤ Approach (Type 2 – Tampines)

Since price has stepwise decrease, we split the resale HDB flats into 2 groups (age group 1-20, 21-60 years). Since σ is unknown, and all age bins have $n > 30$, we performed Z test assuming unequal variances.

- H0: There is no difference in resale price / sqm between the 2 groups
- H1: There is difference in resale price / sqm between the 2 groups

➤ Results

- 3-room: p-value < 5%, we reject H0. Sufficient evidence that mean resale price / sqm differs across the 2 groups
- 4-room: same as above
- 5-room: same as above

➤ **Summary:** Statistical evidence confirms a significant difference in mean resale price / sqm between newer flats (1-20 years) and older flats (21-60 years). (For more details on the price range difference, refer to 95% confidence intervals in the Appendix)

Conclusion: Property companies can use this insight to **develop age-specific pricing strategies**, help their clients to make better decisions on valuation, investment, and timing of resale

5. Impact of Cooling Measures on Resale Prices & Transactions

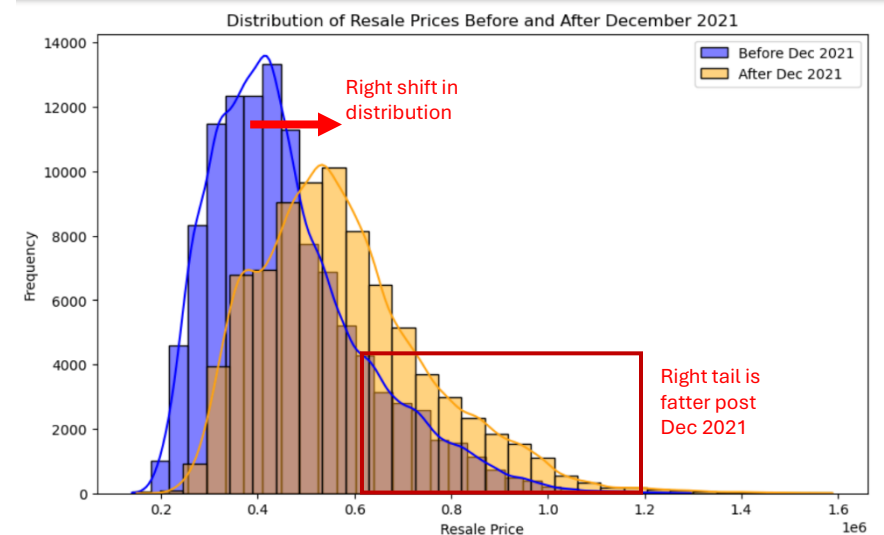
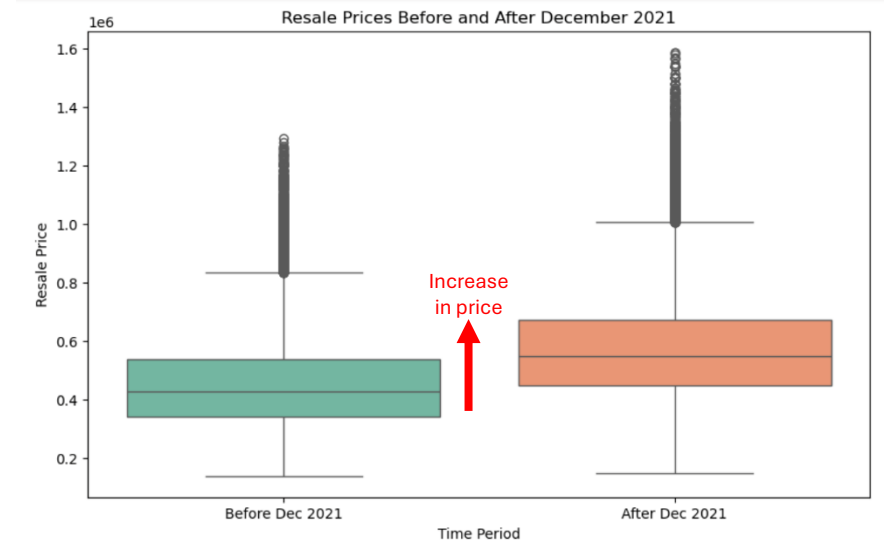
A. Descriptive Statistics

➤ Key Insights:

- **Increased resale prices after Dec 2021*.** There is an upwards shift in the **boxplot** and rightwards shift in **histogram**. The shift suggests that flats were transacted at higher price after post-policy. This may cause resale flats to be less affordable for lower-income buyers.
- **Continued demand for high-value flats remained strong.** From the histogram, we can see that post Dec 2021 (yellow), other than a rightward shift, the right tail of the histogram has become fatter, indicating that buyers are still willing to pay premium prices despite cooling measures.

➤ Strategic Takeaways:

- **For Buyers & Investors:** Purchasing earlier may have helped secure a better deal during COVID times, avoiding further price increases.
- **For Property Companies:** Demand for high-value flats still strong, and there is a lack of affordable lower-cost flats. The company can cater its pricing strategy i.e. target premium buyers for higher-end units, and create incentives or financing options for low-mid tier buyers.
- **For Policymakers:** Decide if more cooling measures are needed to keep housing affordable for lower-income buyers while keeping the market stable.



* Cooling measures were implemented after Dec 2021, so this is used as our cutoff month

5. Impact of Cooling Measures on Resale Prices & Transactions

B. Inferential Statistics

To validate the insights regarding increase in resale price and continued strong demand for high-value flats before vs after cooling measures, we performed hypothesis testings for resale_price and transaction volumes. The data was split into 2 groups – before Dec 2021, and after Dec 2021.

➤ Approach:

- **Test 1:** An independent Z-test was performed to compare the means of resale_price before and after, for each town. Since $n > 30$ for each town-group, σ unknown, a two-tailed Z-test is suitable
 - **H0:** There is no difference in mean resale_price before and after cooling measures
 - **H1:** There is difference in mean resale_price before and after cooling measures
- **Test 2:** Demand for high-value flats is quantified with transaction volume of flat with resale_price ≥ 800000 before and after Dec 2021, and calculated each town, yielding 26 towns * 2 = 52 data points
- As n is small (26 pairs), and transaction volumes before and after are related, an upper-tailed paired t-test was performed to compare the means of transaction volume
 - **H0:** The mean transaction volume after cooling measures is same or less than before
 - **H1:** The mean transaction volume after cooling measures is higher than before

➤ Results:

- **Test 1:** For all towns, **Z-score is high** and **p-value close to 0.0** $< 5\%$ (refer to table on right). We reject H0. Sufficient evidence that mean resale_price before and after cooling measures are different. Specifically, they are higher (from box plot in previous slide)
- **Test 2:** **t-statistic is 4.124** and **p-value is 0.000** $< 5\%$. We reject H0. Sufficient evidence that mean transaction volume is higher after cooling measures.

➤ Conclusion:

- These findings provide strong statistical evidence that the **resale prices have increased** and **demand higher-value flats still increased** after Dec 2021, despite cooling measures. The government may need to roll out stronger cooling measures subsequently.

	Town:	Z-score:	P-value:	Conclusion:
North East	ANG MO KIO	16.717	0.0000	Reject the null hypothesis
	HOUGANG	22.839	0.0000	Reject the null hypothesis
	PUNGGOL	39.593	0.0000	Reject the null hypothesis
	SENGKANG	47.752	0.0000	Reject the null hypothesis
	SERANGOON	11.805	0.0000	Reject the null hypothesis
East	BEDOK	18.017	0.0000	Reject the null hypothesis
	PASIR RIS	25.972	0.0000	Reject the null hypothesis
	TAMPINES	28.566	0.0000	Reject the null hypothesis
Central	BISHAN	10.773	0.0000	Reject the null hypothesis
	BUKIT MERAH	12.275	0.0000	Reject the null hypothesis
	BUKIT TIMAH	2.510	0.0121	Reject the null hypothesis
	CENTRAL AREA	2.782	0.0054	Reject the null hypothesis
	GEYLANG	9.681	0.0000	Reject the null hypothesis
	KALLANG/WHAMPOA	14.116	0.0000	Reject the null hypothesis
	MARINE PARADE	3.078	0.0021	Reject the null hypothesis
	QUEENSTOWN	7.342	0.0000	Reject the null hypothesis
West	TOA PAYOH	10.389	0.0000	Reject the null hypothesis
	BUKIT BATOK	25.189	0.0000	Reject the null hypothesis
	BUKIT PANJANG	22.998	0.0000	Reject the null hypothesis
	CHOA CHU KANG	36.341	0.0000	Reject the null hypothesis
	CLEMENTI	9.885	0.0000	Reject the null hypothesis
	JURONG EAST	10.459	0.0000	Reject the null hypothesis
North	JURONG WEST	31.986	0.0000	Reject the null hypothesis
	SEMBAWANG	29.219	0.0000	Reject the null hypothesis
	WOODLANDS	39.090	0.0000	Reject the null hypothesis
	YISHUN	34.627	0.0000	Reject the null hypothesis

t: 4.124

p-value: 0.000

Mean transaction volume between before and after cooling measures differ

6. Executive Summary

➤ Recap on Problem Statement

The objective of the analysis was to identify buyer demand patterns in the resale HDB market in terms of age, location, and impact by government policies, thus producing actionable insights. The insights in these 3 areas would aid in developing sales strategies, crafting a competitive resale pricing for clients and preparing for future planning

➤ Key Insights & Business Recommendations

	1. High-Growth Towns	2. Flat Age and Resale Price Trends	3. Impact of Cooling Measures
Observations & Key Insights	- <i>Punggol, Sengkang, Bukit Merah, Queenstown, and Hougang</i> lead in both transaction volume and resale prices. - Statistical analysis showed these 5 towns have significantly higher demand, making them prime location for property investment	- Resale prices decline with age, following two patterns: 1. Gradual Decline – Prices drop linearly & 2. Stepwise Decline – Prices remain stable before drops at key age thresholds. - Older flats become less attractive, affecting seller pricing strategies and buyer affordability.	- Prices increased after Dec 2021, shifting towards higher price points. - Fewer lower-priced flats transactions, indicating affordability concerns. - Despite gov cooling measures, demand for high-value flats remained strong.
Recommendation for Property Company	• Making marketing and sales efforts for these 5 towns to capture buyer interest. • Refine pricing strategies based on demand. • Continue to monitor the trends to identify future high-growth locations.	• Advise sellers to list their flats before age threshold to avoid major price drop • Advise buyers to buy flat after major price drops for better affordability. • Tailor pricing strategies for different flat age segments.	• Adjust pricing models base on the trend and demand to reflect affordability shifts. • Diversify offerings by identifying low/mid/high-range segments. • Recommend policymakers with insights on market stability and affordability concerns.

➤ Limitations and Next Steps

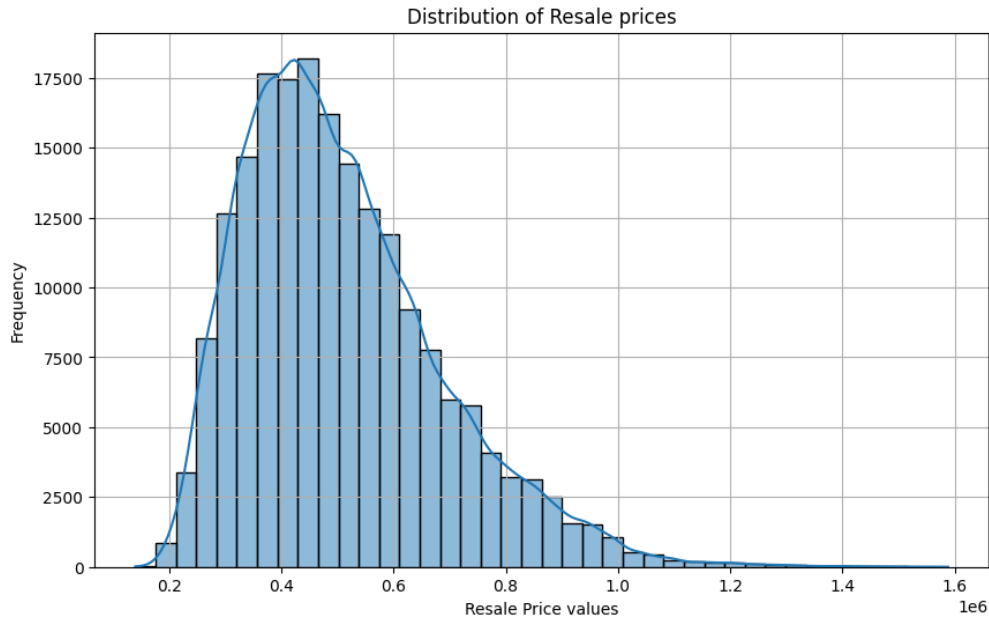
- ANOVA was not allowed as an inferential statistic technique. We therefore did pair-wise hypothesis testing
- We were only given data on HDB resale transactions, which limited the scope of this project. As the real estate market also includes private property (e.g. condominiums, landed), and buyers may be moving between HDB and private property markets, an analysis on private property transactions by towns would also be good to gain a full picture of the trends

Appendix

Appendix – Overview of general trends

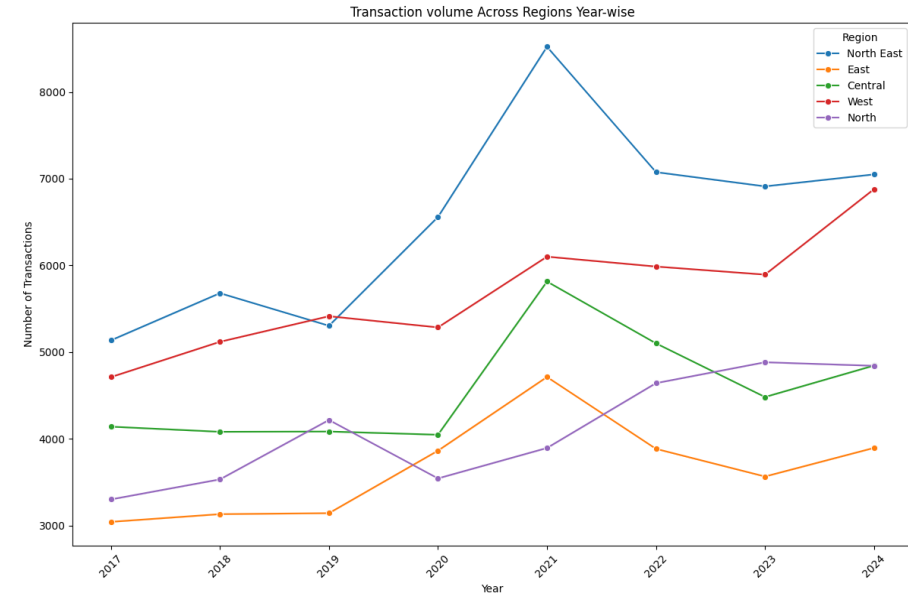
Resale price distribution of housing prices

- The histogram is skewed to the right most resale prices are concentrated on the lower end while fewer properties have very high resale prices.
- The highest frequency occurs between 400,000 and 500,000, indicating that most resale prices fall in this range.

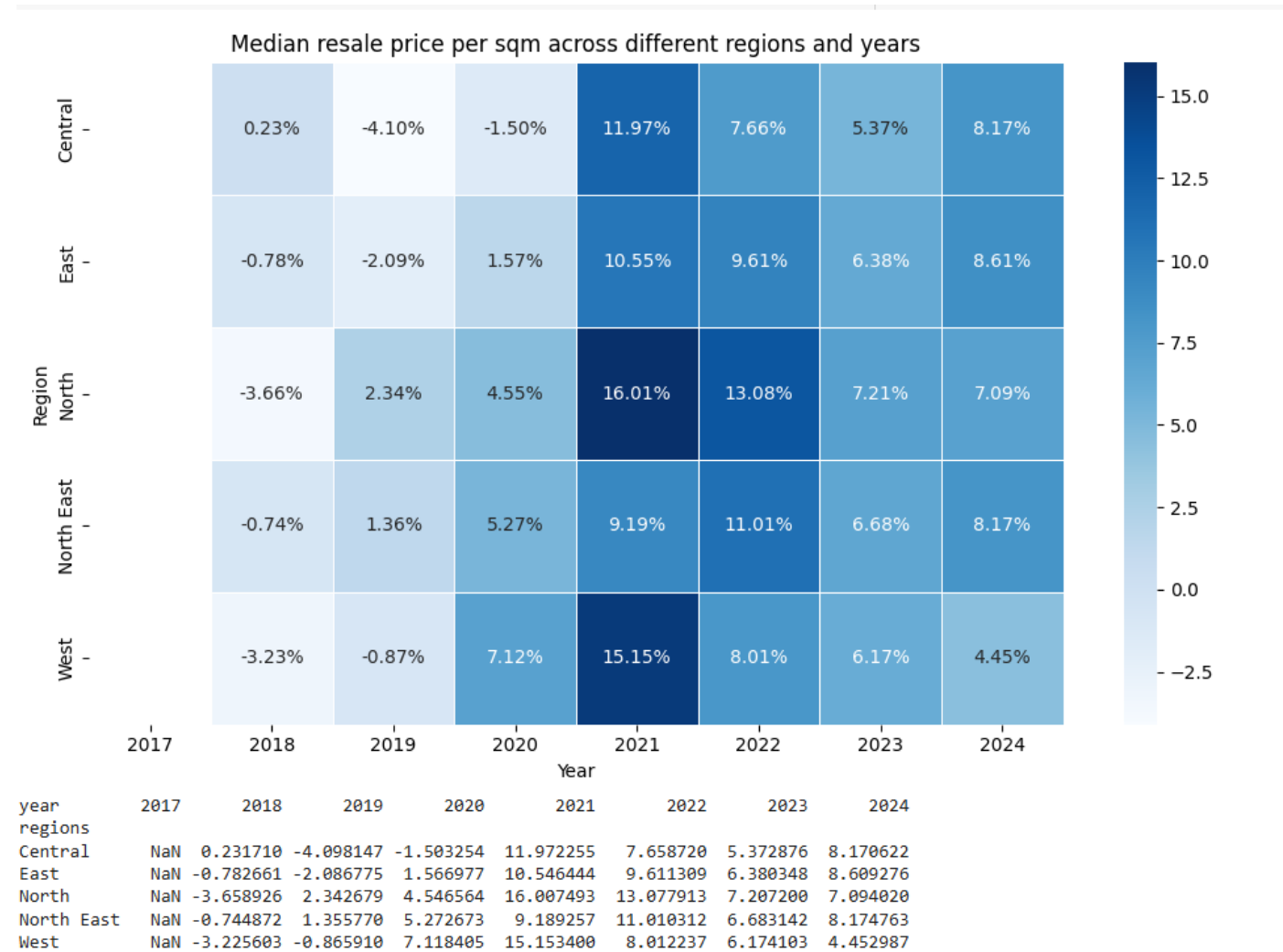


Yearly trends in transaction volume

- The North East region experienced the highest growth rate in transaction volumes from 2019 to 2021
- Although transaction volumes declined after 2021, it still ranks first among the regions in terms of the number of flats resold as of 2024, followed by the West region in second place
- The year 2021 recorded the highest transaction volume across all regions, except for the North region, where transaction volumes began to rise after 2021, surpassing the East and Central regions from 2022

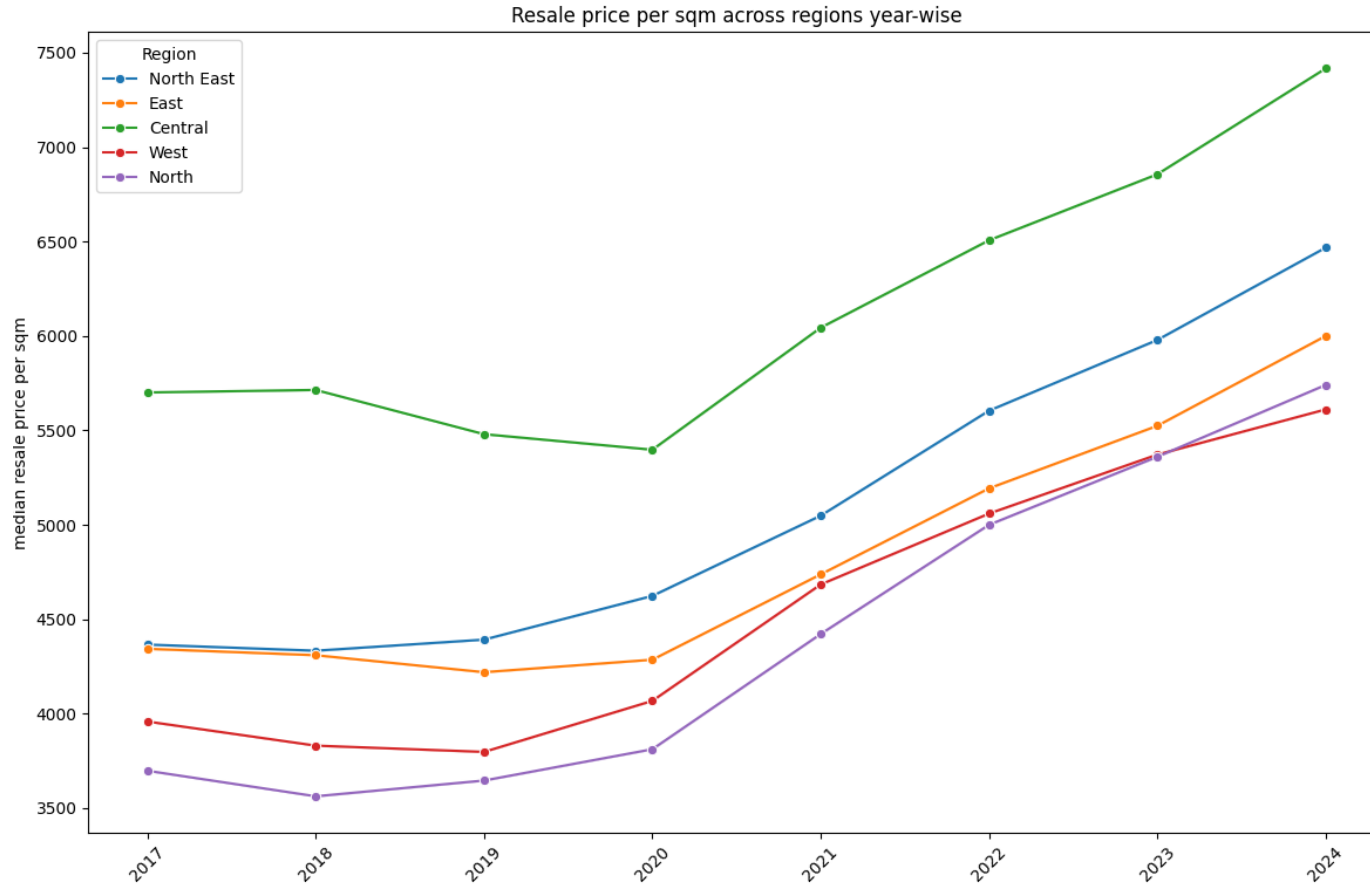


Appendix – Percentage growth in resale housing price



- Most regions show a trend of positive values in the later years (2020-2024), suggesting potential growth or improvement.
- The Central region exhibits the most significant fluctuations, with a large %decrease in 2019 and a large positive % increase in 2021.
- North Region's Strong Growth: The North region shows consistent growth in the past 2 years indicating a strong growth or sign of improvement.
- West Region's Declining Trend: The West region shows a declining trend in the last two years suggesting a potential slowdown or challenges.

Appendix – Yearly price growth line chart



- Central Region: Exhibits the highest price per sqm, particularly after 2020.
- North region starts with the lowest price per square foot, suggesting it offers more affordable housing options. It closely follows the trajectory of East and North East regions.
- East region shows a steady and consistent upward trend, with a noticeable increase from 2020 onwards which ranks 3rd among the other regions in terms of resale pricing per sqm.
- West region demonstrates a similar upward trend to the East and it is trying to catch up with other regions.

Appendix – Rank chart for txn volume and price for 26 towns

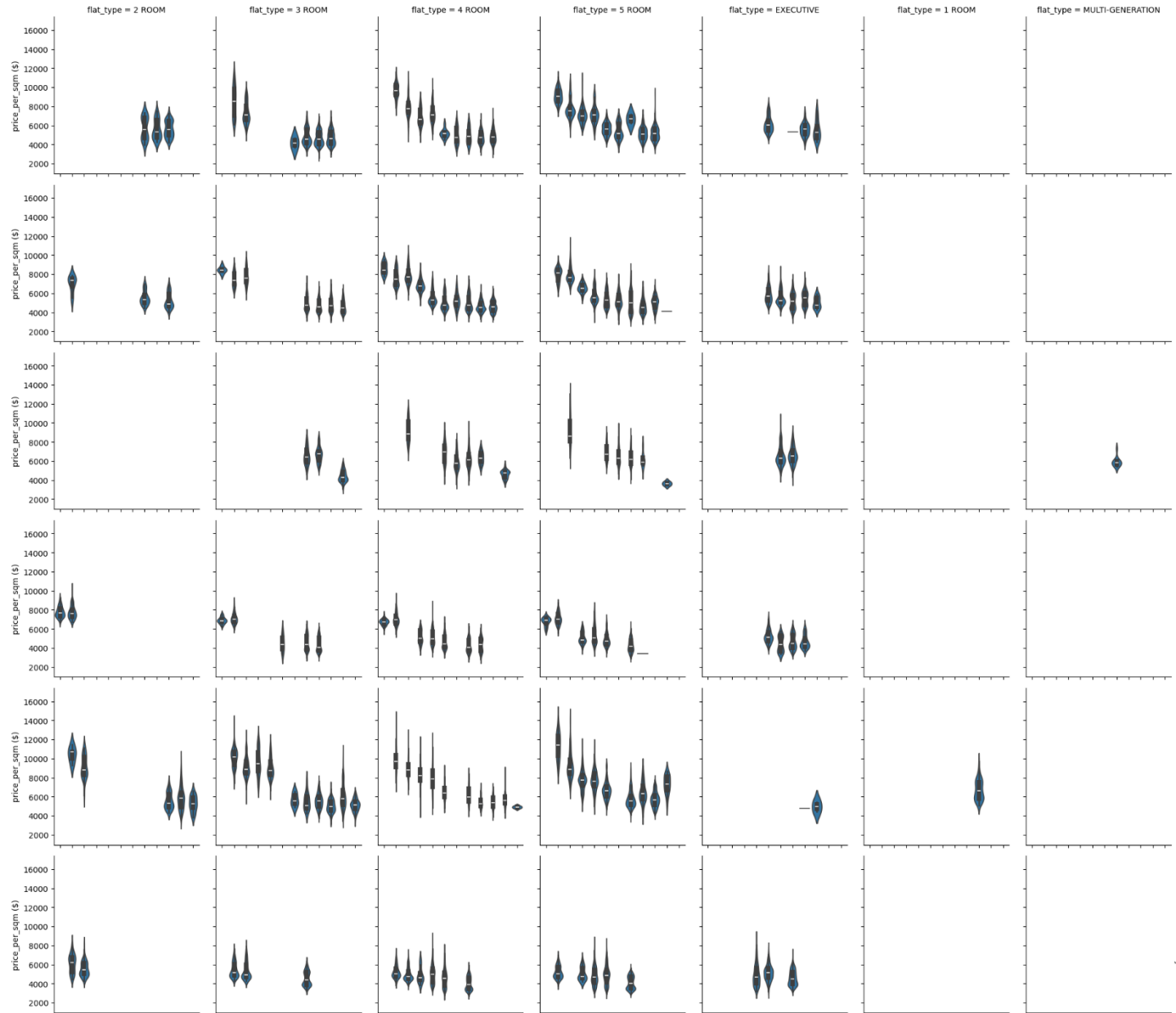
	regions	town	resale_price_sqm
1	Central	CENTRAL AREA	7469.880
2	Central	QUEENSTOWN	7203.390
3	Central	BUKIT MERAH	6652.170
4	Central	BUKIT TIMAH	6635.160
5	Central	BISHAN	6190.480
6	Central	MARINE PARADE	6157.890
7	Central	KALLANG/WHAMPOA	5951.460
8	West	CLEMENTI	5543.480
9	North East	PUNGGOL	5514.710
10	Central	TOA PAYOH	5444.985
11	Central	GEYLANG	5362.045
12	North East	SERANGOON	5349.795
13	North East	SENGKANG	5052.630
14	North East	ANG MO KIO	5000.000
15	East	TAMPINES	4952.380
16	East	BEDOK	4899.495
17	North East	HOUGANG	4886.880
18	West	BUKIT BATOK	4884.620
19	West	BUKIT PANJANG	4730.975
20	North	SEMBAWANG	4704.940
21	West	JURONG EAST	4701.490
22	North	YISHUN	4615.380
23	East	PASIR RIS	4587.300
24	West	CHOA CHU KANG	4455.000
25	West	JURONG WEST	4373.630
26	North	WOODLANDS	4264.710

	regions	town	transaction_volume
1	North East	SENGKANG	108681787
2	North East	PUNGGOL	98941893
3	West	JURONG WEST	74229761
4	North East	HOUGANG	66047404
5	North	WOODLANDS	58266055
6	North	YISHUN	55960428
7	North East	ANG MO KIO	52996435
8	West	CHOA CHU KANG	51878436
9	East	TAMPINES	49271570
10	West	BUKIT BATOK	46069474
11	West	BUKIT PANJANG	40622399
12	East	BEDOK	38312649
13	Central	BUKIT MERAH	34905751
14	Central	TOA PAYOH	28356338
15	Central	KALLANG/WHAMPOA	27925954
16	Central	QUEENSTOWN	25251665
17	West	CLEMENTI	25016725
18	North	SEMBAWANG	23481133
19	North East	SERANGOON	23298778
20	West	JURONG EAST	22972937
21	Central	GEYLANG	22473067
22	East	PASIR RIS	21445714
23	Central	BISHAN	16210043
24	Central	CENTRAL AREA	7250353
25	Central	MARINE PARADE	5603580
26	Central	BUKIT TIMAH	2250821

Appendix – Combined rank score for 26 towns

	regions	town	resale_price_rank	transaction_volume_rank	combined_rank
1	North East	PUNGGOL	9.0	2.0	11.0
2	North East	SENGKANG	13.0	1.0	14.0
3	Central	BUKIT MERAH	3.0	13.0	16.0
4	Central	QUEENSTOWN	2.0	16.0	18.0
5	North East	HOUGANG	17.0	4.0	21.0
6	North East	ANG MO KIO	14.0	7.0	21.0
7	Central	KALLANG/WHAMPOA	7.0	15.0	22.0
8	Central	TOA PAYOH	10.0	14.0	24.0
9	East	TAMPINES	15.0	9.0	24.0
10	West	CLEMENTI	8.0	17.0	25.0
11	Central	CENTRAL AREA	1.0	24.0	25.0
12	Central	BISHAN	5.0	23.0	28.0
13	West	BUKIT BATOK	18.0	10.0	28.0
14	North	YISHUN	22.0	6.0	28.0
15	West	JURONG WEST	25.0	3.0	28.0
16	East	BEDOK	16.0	12.0	28.0
17	West	BUKIT PANJANG	19.0	11.0	30.0
18	Central	BUKIT TIMAH	4.0	26.0	30.0
19	Central	MARINE PARADE	6.0	25.0	31.0
20	North East	SERANGOON	12.0	19.0	31.0
21	North	WOODLANDS	26.0	5.0	31.0
22	Central	GEYLANG	11.0	21.0	32.0
23	West	CHOA CHU KANG	24.0	8.0	32.0
24	North	SEMPAWANG	20.0	18.0	38.0
25	West	JURONG EAST	21.0	20.0	41.0
26	East	PASIR RIS	23.0	22.0	45.0

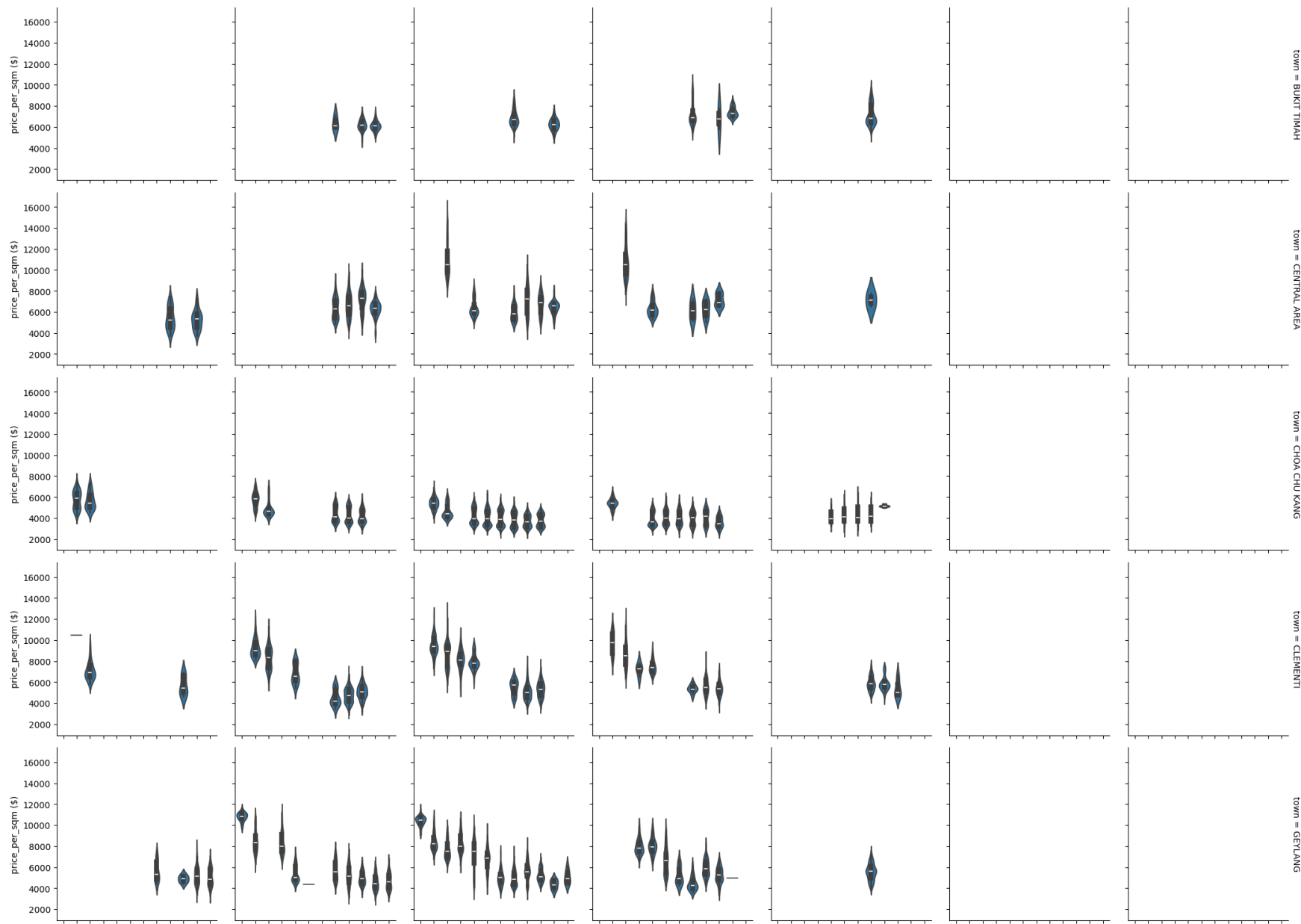
Appendix – Distribution of Flat Age by Town, Flat Type (1/5)



Properties of FacetGrid (Violin Plot)

- Rows: Towns
- Columns: Flat Type (1-5 room, multi gen)
- Sub plot
 - X-axis: Age bins (age_bin)
 - Y-axis: Resale price per sqm (price_per_sqm)

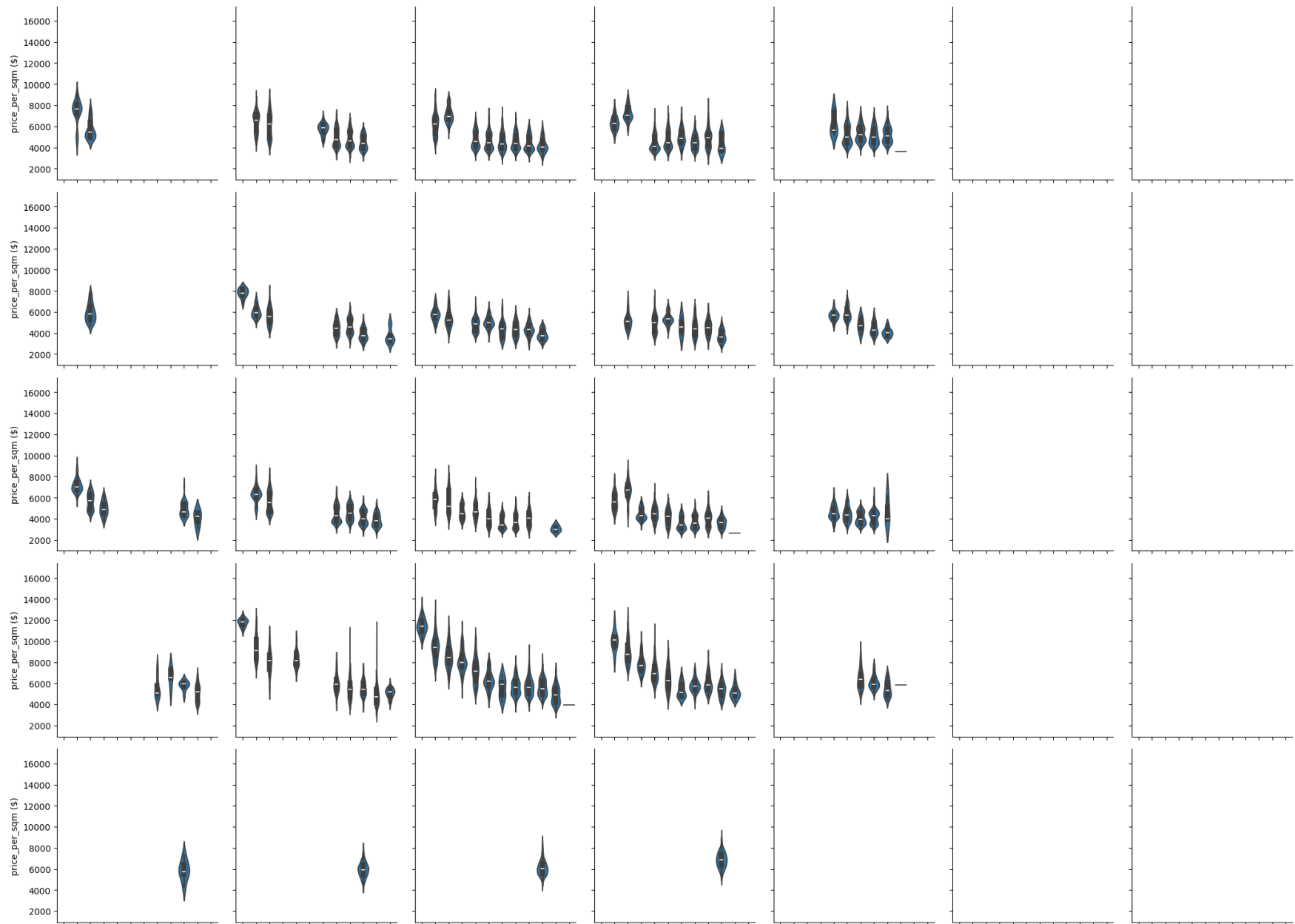
Appendix – Distribution of Flat Age by Town, Flat Type (2/5)



Properties of FacetGrid (Violin Plot)

- Rows: Towns
- Columns: Flat Type (1-5 room, multi gen)
- Sub plot
 - X-axis: Age bins (age_bin)
 - Y-axis: Resale price per sqm (price_per_sqm)

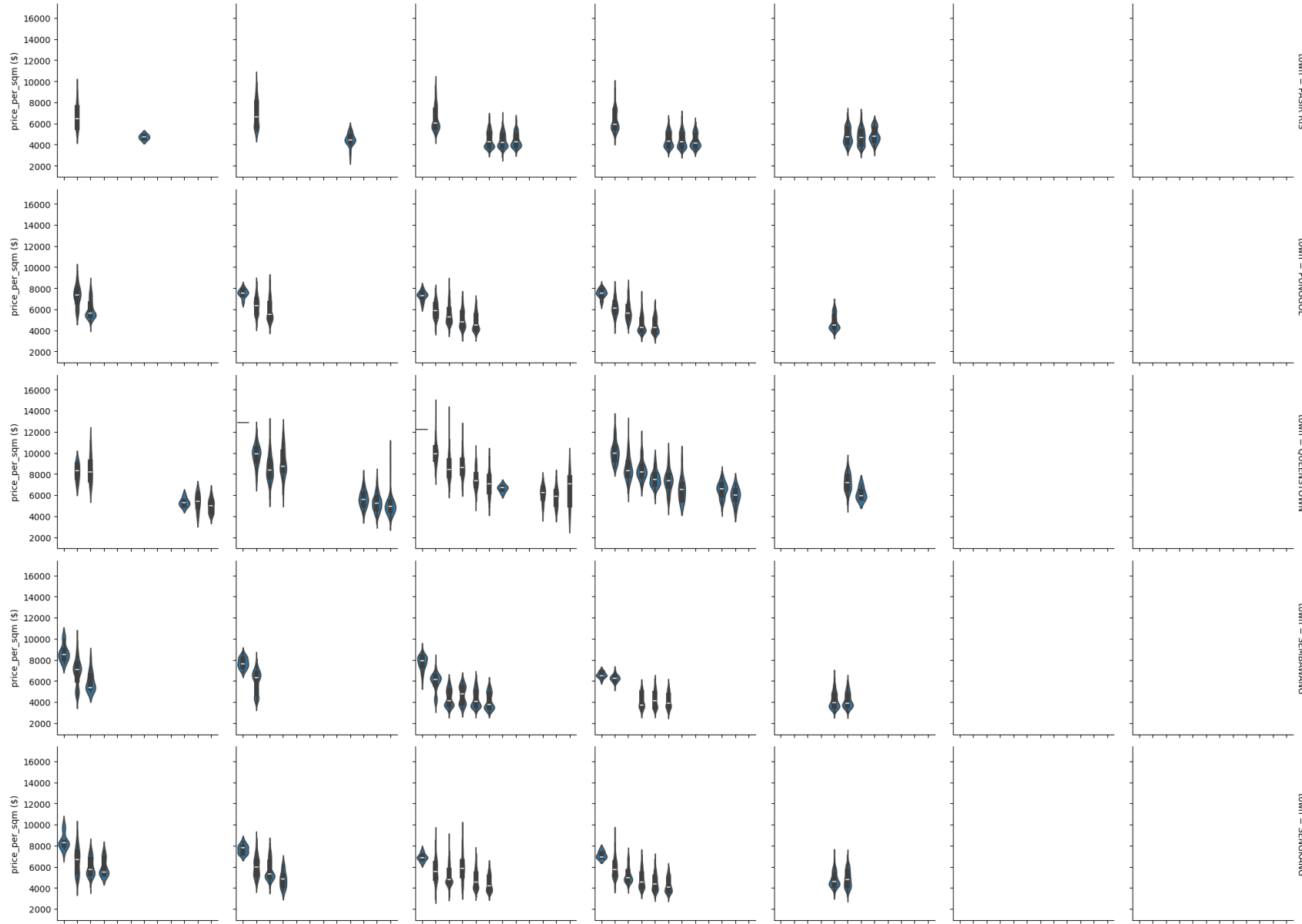
Appendix – Distribution of Flat Age by Town, Flat Type (3/5)



Properties of FacetGrid (Violin Plot)

- Rows: Towns
- Columns: Flat Type (1-5 room, multi gen)
- Sub plot
 - X-axis: Age bins (age_bin)
 - Y-axis: Resale price per sqm (price_per_sqm)

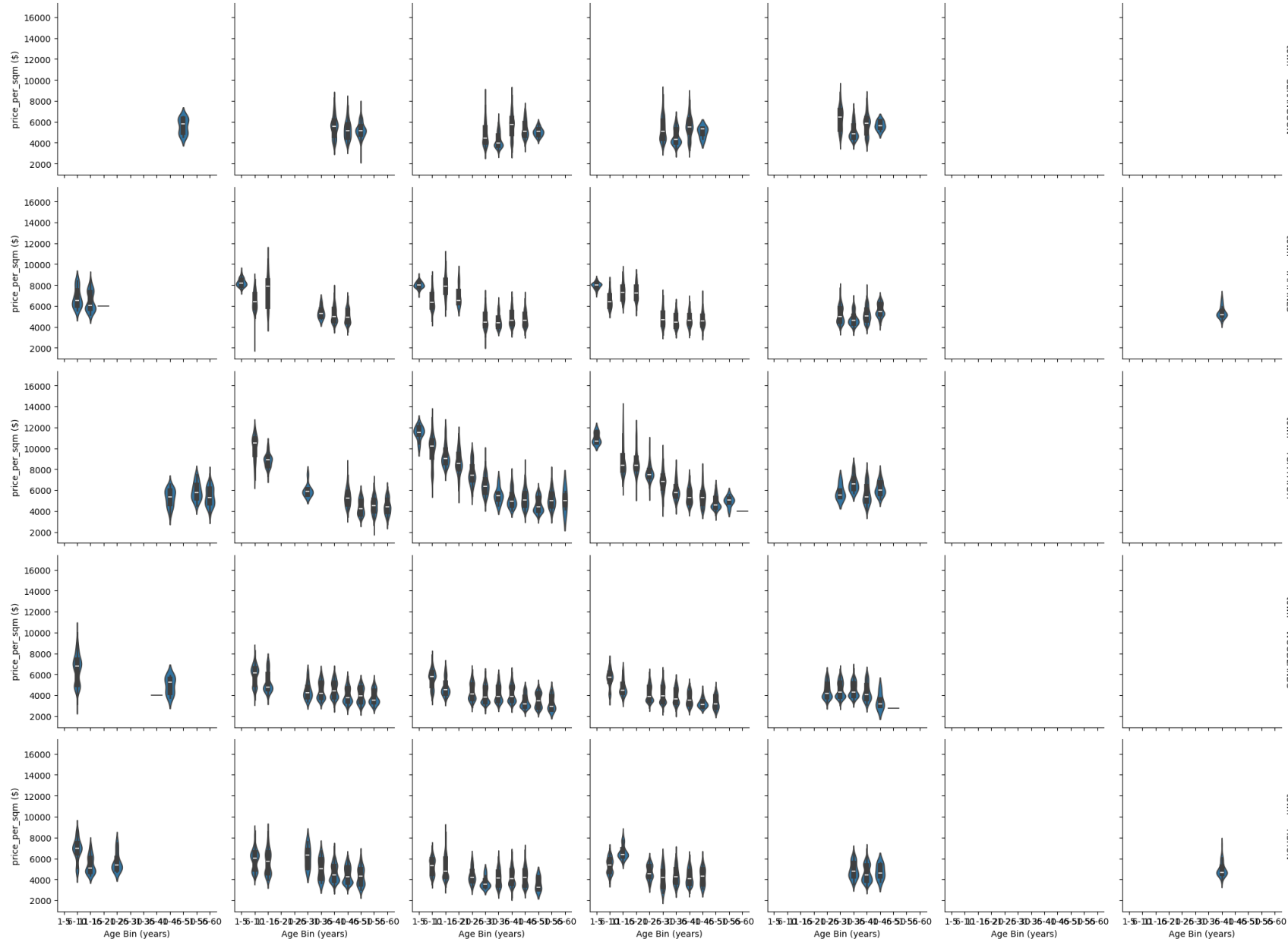
Appendix – Distribution of Flat Age by Town, Flat Type (4/5)



Properties of FacetGrid (Violin Plot)

- Rows: Towns
- Columns: Flat Type (1-5 room, multi gen)
- Sub plot
 - X-axis: Age bins (age_bin)
 - Y-axis: Resale price per sqm (price_per_sqm)

Appendix – Distribution of Flat Age by Town, Flat Type (5/5)



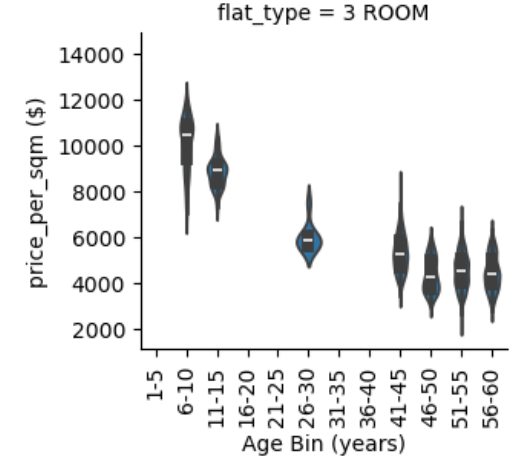
Properties of FacetGrid (Violin Plot)

- Rows: Towns
- Columns: Flat Type (1-5 room, multi gen)
- Sub plot
 - X-axis: Age bins (age_bin)
 - Y-axis: Resale price per sqm (price_per_sqm)

Appendix – Toa Payoh 3 Room t-test result

➤ Toa Payoh 3-room t-test result

- We have two independent populations with at least 1 that has a small sample size and unknown variance/sigma values (with the assumption of non-equal variances).
- For every pair of age bins, we use t-test for two independent populations with $\alpha = 5\%$
 - $H_0: \mu_1 - \mu_2 = 0$ i.e. ($\mu_1 = \mu_2$)
 - $H_1: \mu_1 - \mu_2 \neq 0$ i.e. ($\mu_1 \neq \mu_2$)
- Conclusion:
 - P-value < 5% for almost every pair except age bin no. 10 and 12.
 - There is sufficient evidence to conclude that the age bins differ in resale price per sqm
Resale price per sqm of age bin no. 2 > 3 > 6 > 9 > 10 $\neq$ 11 $\neq$ 12
 - The only exception is age bin no. 10 and 12, where there is insufficient evidence to conclude that the age bins differ in mean resale price per sqm



p-values when testing one age_bin against another							
Age Bin	2	3	6	9	10	11	12
2		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
3	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001
6	<0.001	<0.001		0.01	<0.001	<0.001	<0.001
9	<0.001	<0.001	0.01		<0.001	<0.001	<0.001
10	<0.001	<0.001	<0.001	<0.001		0.01	0.26
11	<0.001	<0.001	<0.001	<0.001	0.01		0.02
12	<0.001	<0.001	<0.001	<0.001	0.26	0.02	

Age Bins:

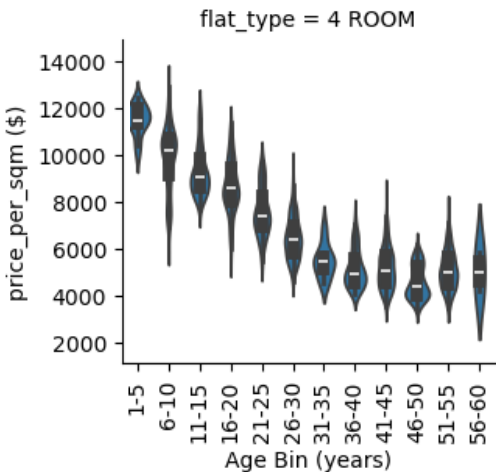
- 1-5
- 6-10
- 11-15
- 16-20
- 21-25
- 26-30
- 31-35
- 36-40
- 41-45
- 46-50
- 51-55
- 56-60

Age Bin (years)	Resale Price Per Sqm (\$)	
	95% CI Lower	95% CI Upper
6-10	9925	10316
11-15	8648	9054
26-30	5517	6451
41-45	5248	5401
46-50	4242	4501
51-55	4506	4585
56-60	4388	4523

Appendix – Toa Payoh 4 Room t-test result

➤ Toa Payoh 4-room t-test result

- We have two independent populations with at least 1 that has a small sample size and unknown variance/sigma values (with the assumption of non-equal variances).
- For every pair of age bins, we use t-test for two independent populations with $\alpha = 5\%$
 - $H_0: \mu_1 - \mu_2 = 0$ i.e. ($\mu_1 = \mu_2$)
 - $H_1: \mu_1 - \mu_2 \neq 0$ i.e. ($\mu_1 \neq \mu_2$)
- Conclusion:
 - P-value < 5% for almost every pair except age bins no. 8, 9, 11, 12
 - There is sufficient evidence to conclude that the age bins differ in resale price per sqm
Resale price per sqm of age bin no. 1 > 2 > 3 > 4 > 5 > 6 > 7 > 8, 9, 10, 11, 12
 - The exception is age bins no. 8, 9, 11, 12, where there is insufficient evidence to conclude that the age bins differ in mean resale price per sqm



p-values when testing one age_bin against another												
Age Bin	1	2	3	4	5	6	7*	8	9	10	11	12*
1		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.07
2	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.10
3	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.12
4	<0.001	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.14
5	<0.001	<0.001	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.20
6	<0.001	<0.001	<0.001	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	0.33
7	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001		0.14	0.17	<0.001	0.09	0.68
8	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.14		0.73	<0.001	0.64	0.90
9	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.17	0.73		<0.001	0.34	0.88
10	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001		<0.001	0.69
11	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.09	0.64	0.34	<0.001		0.94
12	0.07	0.10	0.12	0.14	0.20	0.33	0.68	0.90	0.88	0.69	0.94	

Age Bin 7 and 12 have small sample sizes (13 and 2 respectively)

Age Bins:
1. 1-5
2. 6-10
3. 11-15
4. 16-20
5. 21-25
6. 26-30
7. 31-35
8. 36-40
9. 41-45
10. 46-50
11. 51-55
12. 56-60

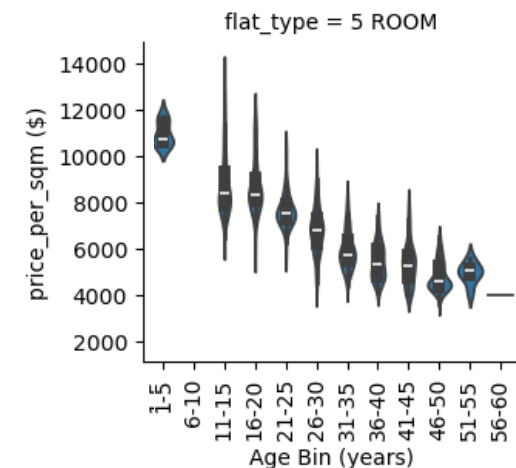
Age Bin (years)	Resale Price Per Sqm (\$)	
	95% CI Lower	95% CI Upper
1-5	11199	11810
6-10	9715	10074
11-15	9121	9431
16-20	8574	8802
21-25	7393	7789
26-30	6361	6565
31-35	5047	5904
36-40	4993	5302
41-45	5073	5288
46-50	4482	4678
51-55	4990	5217
56-60**	N.A.	N.A.

** No confidence interval due to small sample size

Appendix – Toa Payoh 5 Room t-test result

➤ Toa Payoh 5-room t-test result

- We have two independent populations with at least 1 that has a small sample size and unknown variance/sigma values (with the assumption of non-equal variances).
- For every pair of age bins, we use t-test for two independent populations with $\alpha = 5\%$
 - $H_0: \mu_1 - \mu_2 = 0$ i.e. ($\mu_1 = \mu_2$)
 - $H_1: \mu_1 - \mu_2 \neq 0$ i.e. ($\mu_1 \neq \mu_2$)
- Conclusion:
 - P-value < 5% for almost every pair except age bins no. 3 & 4, 8 & 9, 9 & 11, 12 (N.A.)
 - There is sufficient evidence to conclude that the age bins differ in resale price per sqm
Resale price per sqm of age bin no. 1 > 3, 4 > 5 > 6 > 7 > 8, 9 > 10, 11, 12
 - The exception is age bins no. 8 & 9, 9 & 11, 12 (N.A.), where there is insufficient evidence to conclude that the age bins differ in mean resale price per sqm



p-values when testing one age_bin against another											
Age Bin	1	3	4	5	6	7	8	9	10	11	12*
1		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	NaN
3	<0.001		0.41	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	NaN
4	<0.001	0.41		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	NaN
5	<0.001	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	NaN
6	<0.001	<0.001	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	<0.001	NaN
7	<0.001	<0.001	<0.001	<0.001	<0.001		<0.001	<0.001	<0.001	<0.001	NaN
8	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001		0.32	<0.001	0.03	NaN
9	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.32		<0.001	0.08	NaN
10	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001		0.47	NaN
11	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.03	0.08	0.47		NaN
12	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

Age Bins:
1. 1-5
2. 6-10
3. 11-15
4. 16-20
5. 21-25
6. 26-30
7. 31-35
8. 36-40
9. 41-45
10. 46-50
11. 51-55
12. 56-60

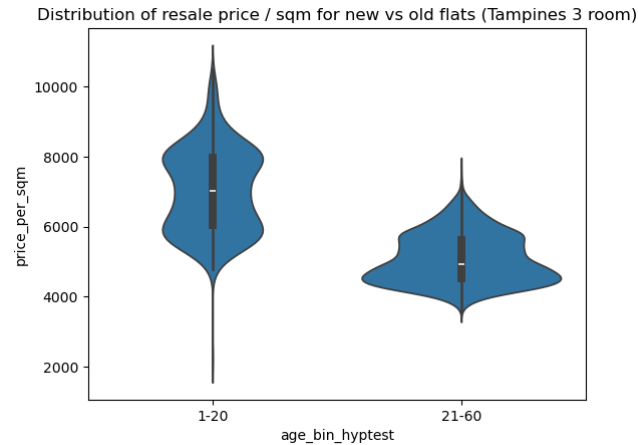
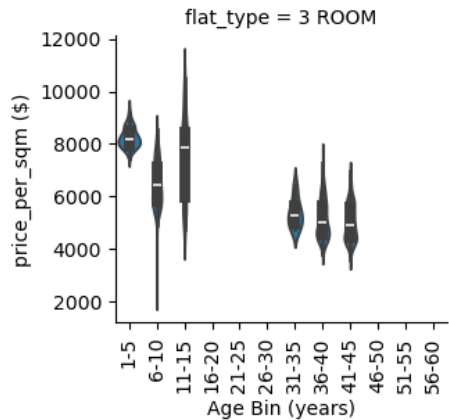
Age Bin (years)	Resale Price Per Sqm (\$)	
	95% CI Lower	95% CI Upper
1-5	10540	11406
11-15	8666	8982
16-20	8401	8975
21-25	7497	7849
26-30	6684	6926
31-35	5775	6130
36-40	5309	5604
41-45	5174	5513
46-50	4706	4927
51-55	4532	5380
56-60*	NaN	NaN

Age Bin 12 only has 1 sample so unable to apply the tests

Appendix – Tampines 3 Room Z-test result

➤ Tampines 3-room Z-test result

- We have two independent populations (Age Grp 1: 1-20 and Age Grp 2: 21-60) with large sample size (Grp 1: 477; Grp 2: 2320) and unknown variance/sigma values (with the assumption of non-equal variances).
- We use Z-test for two independent populations with sigma estimated with s, and $\alpha = 5\%$
 - $H_0: \mu_1 - \mu_2 = 0$ i.e. ($\mu_1 = \mu_2$)
 - $H_1: \mu_1 - \mu_2 \neq 0$ i.e. ($\mu_1 \neq \mu_2$)
- Results:
 - z_score: 34.381
 - critical_value: 1.960
 - p-value: 0.000 (3 d.p.)
- Conclusion:
 - P-value < 5%. There is sufficient evidence to conclude that the 2 groups differ in mean resale price per sqm

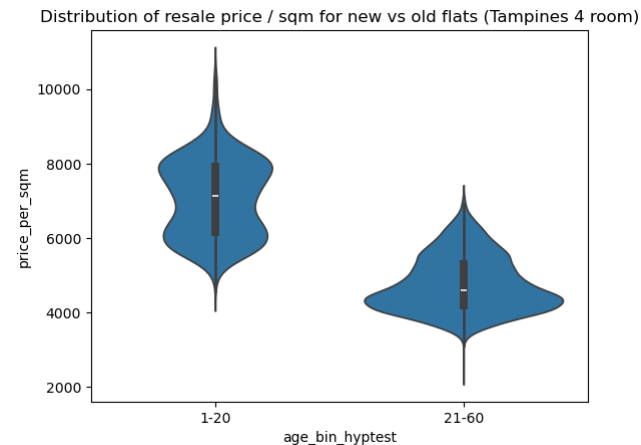
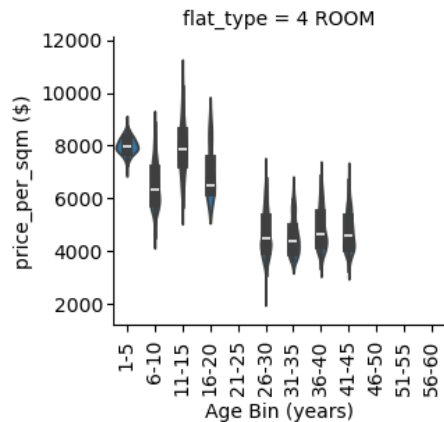


	Resale Price Per Sqm (\$)	
Age Bin (years)	95% CI Lower	95% CI Upper
1-20	6921	7135
21-60	5054	5111

Appendix – Tampines 4 Room Z-test result

➤ Tampines 4-room Z-test result

- We have two independent populations (Age Grp 1: 1-20 and Age Grp 2: 21-60) with large sample size (Grp 1: 969; Grp 2: 4374) and unknown variance/sigma values (with the assumption of non-equal variances).
- We use Z-test for two independent populations with sigma estimated with s, and $\alpha = 5\%$
 - $H_0: \mu_1 - \mu_2 = 0$ i.e. ($\mu_1 = \mu_2$)
 - $H_1: \mu_1 - \mu_2 \neq 0$ i.e. ($\mu_1 \neq \mu_2$)
- Results:
 - z_score: 65.843
 - critical_value: 1.960
 - p-value: 0.000 (3 d.p.)
- Conclusion:
 - P-value < 5%. There is sufficient evidence to conclude that the 2 groups differ in mean resale price per sqm

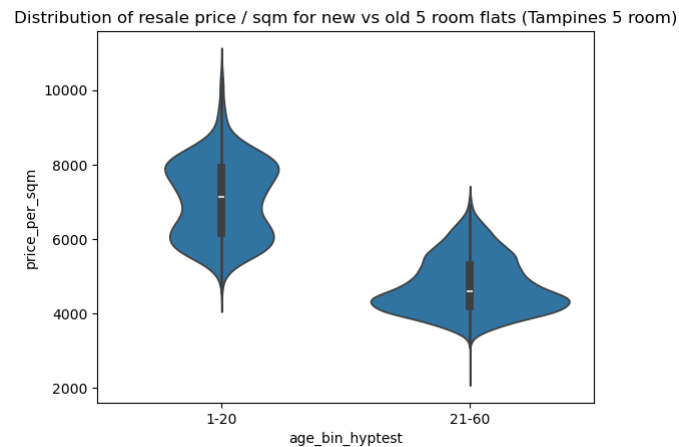
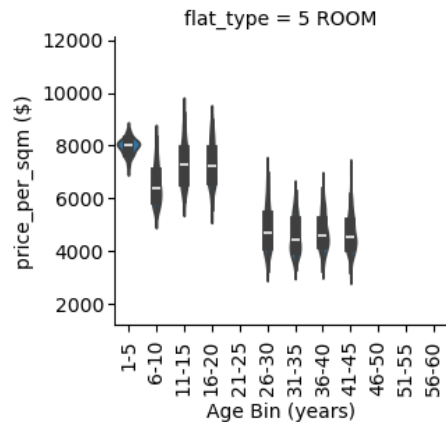


	Resale Price Per Sqm (\$)	
Age Bin (years)	95% CI Lower	95% CI Upper
1-20	7008	7139
21-60	4731	4775

Appendix – Tampines 5 Room Z-test result

➤ Tampines 5-room Z-test result

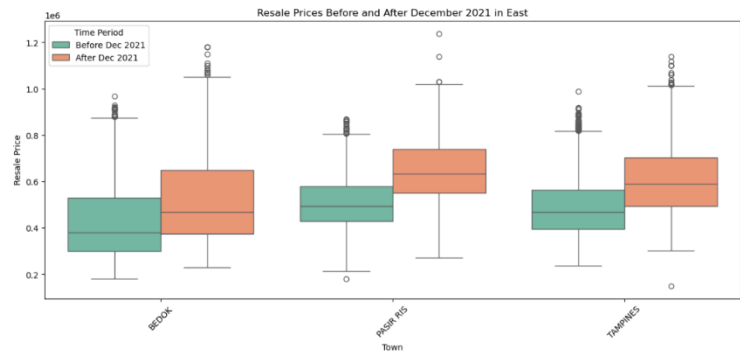
- We have two independent populations (Age Grp 1: 1-20 and Age Grp 2: 21-60) with large sample size (Grp 1: 633; Grp 2: 2915) and unknown variance/sigma values (with the assumption of non-equal variances).
- We use Z-test for two independent populations with sigma estimated with s, and $\alpha = 5\%$
 - $H_0: \mu_1 - \mu_2 = 0$ i.e. ($\mu_1 = \mu_2$)
 - $H_1: \mu_1 - \mu_2 \neq 0$ i.e. ($\mu_1 \neq \mu_2$)
- Results:
 - z_score: 65.680
 - critical_value: 1.960
 - p-value: 0.000 (3 d.p.)
- Conclusion:
 - P-value < 5%. There is sufficient evidence to conclude that the 2 groups differ in mean resale price per sqm



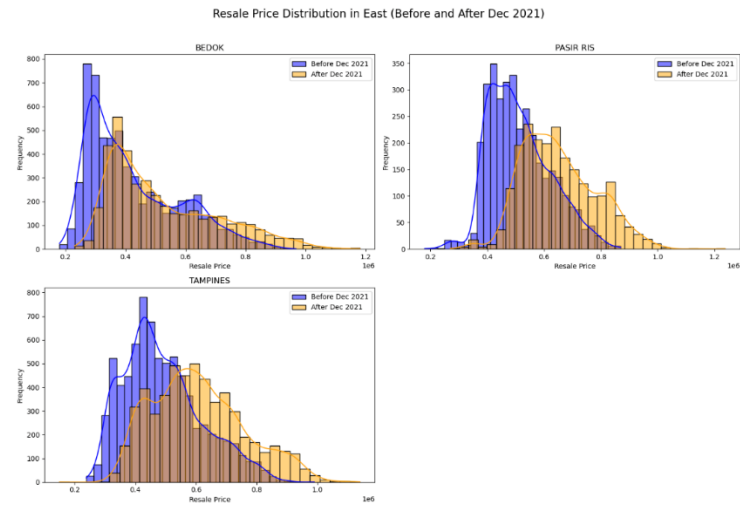
	Resale Price Per Sqm (\$)	
Age Bin (years)	95% CI Lower	95% CI Upper
1-20	7012	7145
21-60	4681	4731

Appendix – Impact of Cooling Measures by Region (1/2)

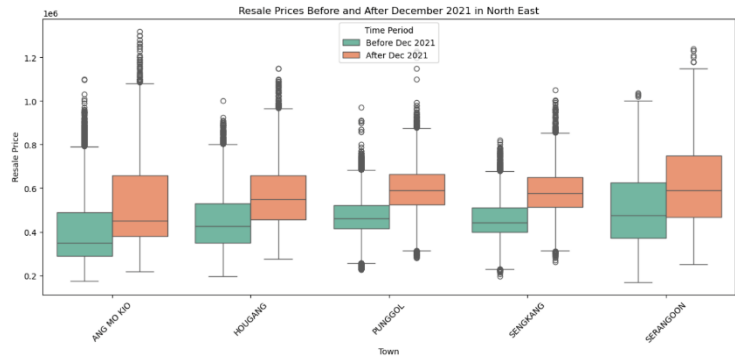
Resale Price before and after Dec 2021 - Boxplots (East)



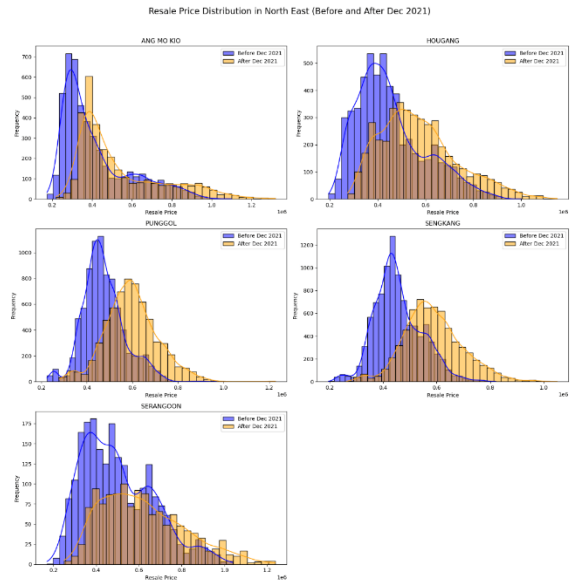
Resale Price distribution before and after Dec 2021 - Histograms (East)



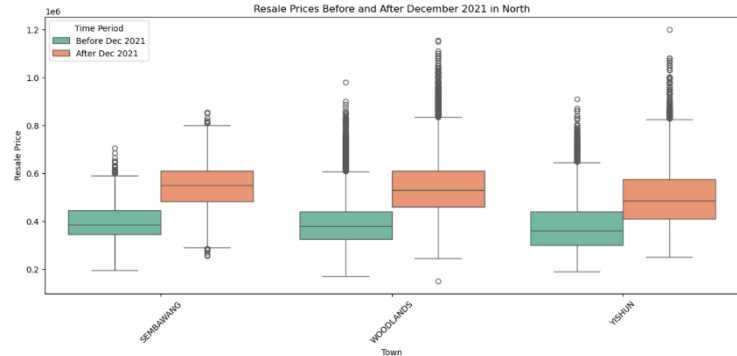
Resale Price before and after Dec 2021 - Boxplots (North East)



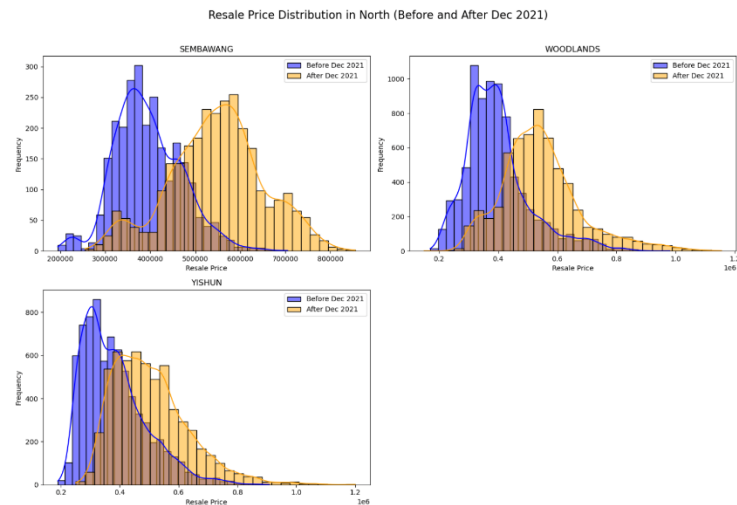
Resale Price distribution before and after Dec 2021 - Histograms (North East)



Resale Price before and after Dec 2021 - Boxplots (North)

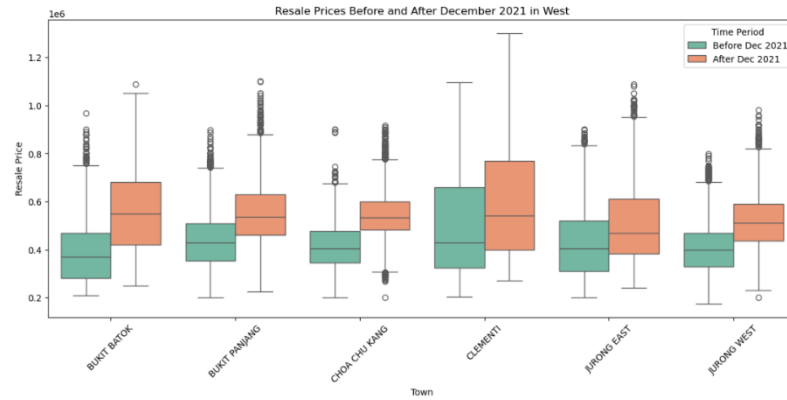


Resale Price distribution before and after Dec 2021 - Histograms (North)

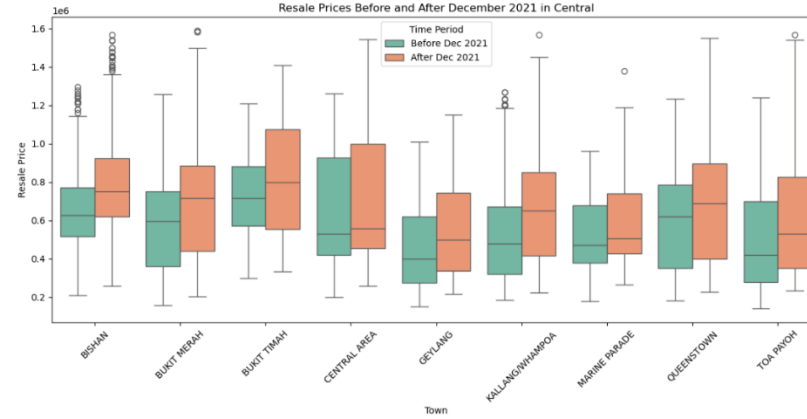


Appendix – Impact of Cooling Measures by Region (2/2)

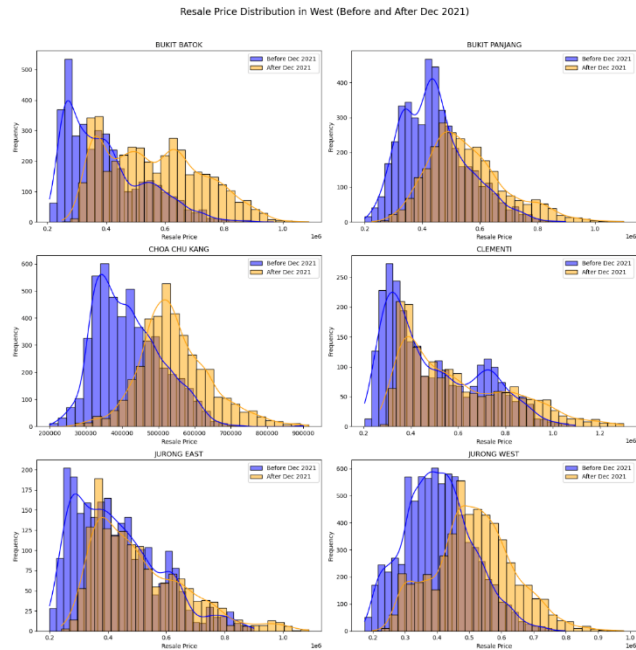
Resale Price before and after Dec 2021 - Boxplots (West)



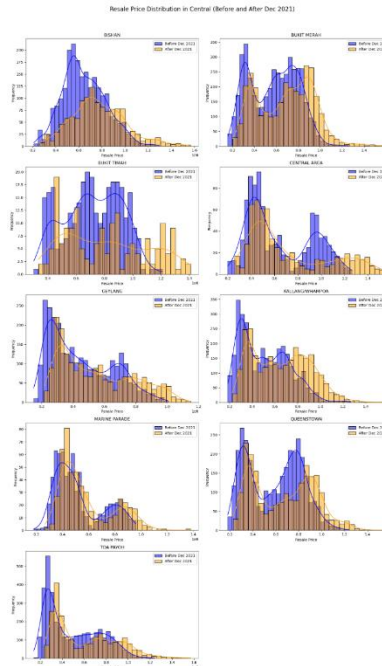
Resale Price before and after Dec 2021 - Boxplots (Central)



Resale Price distribution before and after Dec 2021 - Histograms (West)



Resale Price distribution before and after Dec 2021 - Histograms (Central)



Appendix – Information on Cooling Measures

- **Cooling measures or factors affecting prices rise after 2021** (Ministry of National Development, 2021)
- With effect from 16 December 2021, Additional Buyer's Stamp Duty (ABSD) rates will be raised, and the Total Debt Servicing Ratio (TDSR) threshold will be tightened.
 - The Government will also tighten the LTV limit for loans from HDB from 90% to 85%.
 - Given there is a demand after COVID-19 for families to relocate to a bigger housing unit.
 - Hence, we would like to find out if the 'cooling measures' will have an effect on resale pricing after December 2021.

Therefore, we have chosen Dec 2021 as our cut-off date to split our data into 2 periods: **before** and **after** cooling measures